

Habitat Differences Influence Genetic Impacts of Human Land Use on the American Beech (*Fagus grandifolia*)

CANDICE Y. LUMIBAO AND JASON S. MCLACHLAN

From the Department of Biological Sciences, University of Notre Dame, 294 Galvin Life Sciences, Notre Dame, IN 46556 (Lumibao); and the Department of Biological Sciences, University of Notre Dame, Notre Dame, IN (McLachlan).

Address correspondence to Candice Y. Lumibao at the address above, or e-mail: canlumibao@gmail.com.

Data deposited at Dryad: <http://dx.doi.org/doi:10.5061/dryad.c1q38>

Abstract

Natural reforestation after regional forest clearance is a globally common land-use sequence. The genetic recovery of tree populations in these recolonized forests may depend on the biogeographic setting of the landscape, for instance whether they are in the core or in the marginal part of the species' range. Using data from 501 individuals genotyped across 7 microsatellites, we investigated whether regional differences in habitat quality affected the recovery of genetic variation in a wind-pollinated tree species, American beech (*Fagus grandifolia*) in Massachusetts. We compared populations in forests that were recolonized following agricultural abandonment to those in remnant forests that have only been logged in both central inland and marginal coastal regions. Across all populations in our entire study region, recolonized forests showed limited reduction of genetic diversity as only observed heterozygosity was significantly reduced in these forests ($H_O = 0.520$ and 0.590 , respectively). Within inland region, this pattern was observed, whereas in the coast, recolonized populations exhibited no reduction in all genetic diversity estimates. However, genetic differentiation among recolonized populations in marginal coastal habitat increased (F_{st} logged = 0.072 ; F_{st} secondary = 0.249), with populations showing strong genetic structure, in contrast to inland region. These results indicate that the magnitude of recovery of genetic variation in recolonized populations can vary at different habitats.

Subject areas: Conservation genetics and biodiversity; Population structure and phylogeography

Key words: *Fagus grandifolia*, forest fragmentation, genetic diversity, genetic structure, land use, New England

A primary concern in forest conservation genetics is the potential negative impact of human land use on genetic diversity in trees. Given the concerns about forest fragmentation, it is natural that many studies are concerned with potential reductions in diversity in remnant patches of formerly continuous forests (e.g., [Jump and Penuelas 2006](#); [Dubreuil et al. 2010](#)). In addition to forest fragmentation, however, natural forest regeneration is an ongoing and common feature in many regions of the world, including eastern North America and Europe ([Foster 2002](#); [Chauchard et al. 2007](#); [Prishchepov et al. 2013](#)). Fewer studies focus on the potential recovery of genetic diversity in forests that have naturally regenerated (but see [Sezen et al. 2007](#); [Kitamura et al. 2008](#)). It is well documented that species diversity recovers slowly in reforesting landscape ([Leahy and Pregitzer 2003](#); [Hall et al. 2004](#)). The recovery of genetic diversity in these regenerated forests may be equally slow.

In general, however, forest trees are considered resistant to the negative effects of deforestation and fragmentation due to their life-history attributes, for example, longevity, high heterozygosity, and extensive gene flow by pollen and seeds ([Hamrick 2004](#)). In studies of continuous forests surviving land-use fragmentation (remnant forests), negative effects of fragmentation occur primarily in small populations isolated for long time periods (e.g., [Jump and Penuelas 2006](#); [Leonardi et al. 2012](#)). These include genetic bottlenecks, reduced genetic diversity, and, in some cases, altered genetic structure ([Fore et al. 1992](#); [Jump and Penuelas 2006](#); [Dubreuil et al. 2010](#)). The genetic characteristics of these “remnant” forests influences the genetic recovery of “recolonized forests” (cleared land that is subsequently reforested, herein referred to as recolonized forests) because trees in recolonized forests descend from ancestors in remnant forests. For instance, genetic diversity can decrease, increase, or

be maintained in recolonized forests depending on whether founder individuals come from multiple genetically diverse sources or from fewer genetically depauperate local populations (Cespedes et al. 2003; Lesser et al. 2013). Similarly, population differentiation in recolonized forests can either increase or decrease depending on the levels of gene flow via seed and pollen flow among populations (Sork et al. 1999; Sezen et al. 2007). Seed dispersal and seedling establishment at suitable sites also influence genetic recovery as they affect effective gene flow in recolonized forests and the number of founder individuals (Gillman 1997; Ouborg et al. 1999).

The processes described above may be particularly pronounced in marginal habitats. For instance, peripheral habitats often contain small, spatially disjunct populations due to lower suitability of the environment. As a consequence, peripheral populations may harbor lower genetic diversity and increased genetic differentiation due to low gene flow relative to populations at the core of the species distribution (Lesica and Allendorf 1995; Lammi et al. 1999; Leimu et al. 2006), though this is not always the case (Eckert et al. 2008; Sexton et al. 2009). Reductions in genetic diversity may also occur at range margins as a result of longer term phylogeographic history such as postglacial migration (Hewitt 2004). In combination, we expect that the genetic recovery of recolonized forests will be most difficult in marginal habitats and in cases where the logged remnant forests are small and isolated.

Here, we examine the extent to which levels and patterns of genetic variation was altered in recolonized populations of American beech (*Fagus grandifolia*), a late-successional tree species, along a gradient of environmental suitability in Massachusetts, USA. The region was nearly completely forested at the time of European settlement (400 years ago), but agricultural clearance eliminated almost 80% of these forests by the early 19th century (Foster 1992). Subsequent natural reforestation produced a landscape where forests cover almost their full original extent (Foster 1992; Eberhardt et al. 2003). Historical reconstructions suggest that the census population size of American beech, like those of other disturbance-sensitive species, has not recovered its original (pre-European settlement) abundance after regional forest clearance (Foster et al. 1998; Fuller et al. 1998, 2002). We examined the effect of habitat differences between core (central) and marginal (coastal) populations on the genetic recovery of beech in recolonized forests compared with remnant forests. In Massachusetts, these remnant forests are likely repeatedly logged (herein referred to as logged forests) although they may still represent genetic variation of original undisturbed beech forests as beech is a prolific sprouter. We first tested the hypothesis (hypothesis 1) that logged populations in the coast have lower genetic diversity and increased genetic differentiation than the central, inland populations (central–marginal hypothesis) to examine distribution of genetic variation in these logged populations as they are the sources of recolonizing individuals in recolonized forests. Then, we tested if (hypothesis 2) recolonized forests exhibit reduced genetic diversity and increased genetic differentiation compared with logged forests; an alternative hypothesis

is that recolonized forests will exhibit increased genetic diversity compared with logged populations. Finally, we tested whether (hypothesis 3) the magnitude of reduction in genetic diversity in recolonized forests compared with logged forests is greater at marginal than at central habitats.

Materials and Methods

Study Region and Species

Our study area is located in Massachusetts in the northeastern United States (Figure 1A). It spans a soil and climate gradient from the coast to the interior, which influences the distribution of many tree species across the region, including American beech (Cogbill et al. 2002). The higher elevation inland region has a cooler and wetter climate than the coastal region. It is dominated by mesic soils, whereas the coast is dominated by sandy outwash soils with islands of more mesic morrainal till (Foster 2002).

American beech is a wind-pollinated and late-successional tree species common in northeastern North America. It is shade-tolerant, and juveniles can survive more than 70 years of shade suppression (Poulson and Platt 1989; Canham 1990). Beech seeds are relatively heavy and dispersed by rodents and bluejays (Johnson and Adkisson 1985). Beech has 2 modes of reproduction: sexually by seed and asexually by root suckers (Ward 1961; Kitamura and Kawano 2001).

The postglacial history of American beech in New England has been studied previously using paleoecological, historical, and genetic data (Davis 1981; McLachlan et al. 2005). Beech arrived around 10 000 years ago in Massachusetts and became abundant ~8000–6000 years ago during a period of climate warming (Davis 1981; Webb 1987; Foster et al. 2002). Throughout the Holocene, beech was more common in mesic inland sites than on the coast. At the time of European settlement, beech was most abundant in northwest and central Massachusetts but irregularly distributed in low abundance in southern and coastal region (Figure 1A). Most coastal beech stands are <5 ha and restricted to mesic soils on moraines (Busby et al. 2009). Following European settlement, forests in Massachusetts were extensively logged and cleared for agriculture and pasture. At the peak of forest clearance in the early 1800s, almost 80% of inland Massachusetts and ~44% of coastal region was cleared for agriculture or pasture (Figure 1B) (Foster 1992; Eberhardt et al. 2003). Remnant forest fragments generally remained forested throughout postsettlement period, though they experienced repeated logging. These logged forests, thus, comprise original beech trees, stem sprouts, and postsettlement recruits, likely derived from parents within the fragment. Subsequent abandonment of open agricultural land in the late 1800s resulted in natural regeneration of “secondary or recolonized forests” (Leahy and Pregitzer 2003; Hall et al. 2004). Beech trees in recolonized forests are all derived from seed sources from logged forests. Beech bark disease, a scale-fungus disease complex, has affected these beech forests since its spread into Massachusetts ~60 years ago (Houston et al. 1979) although it has not significantly altered forest structure and composition (Busby 2006).

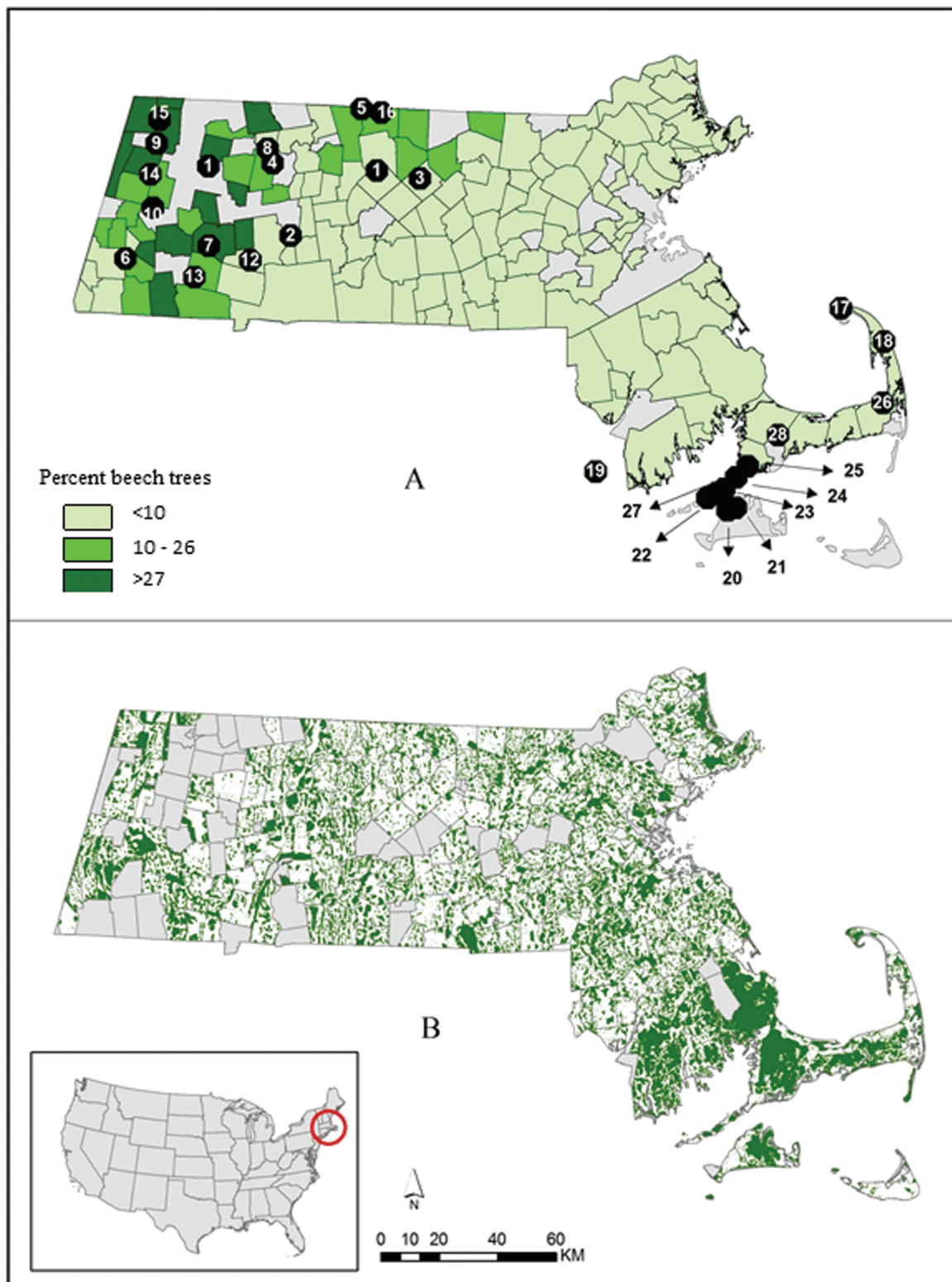


Figure 1. (A) Distribution of American beech reconstructed from township tree surveys at time of European settlement in Massachusetts. Beech abundance is higher in inland region than in the coast (darker shade of green). Circles denote our population location. (B) Forest cover at the peak of forest clearance in the year 1830 shows a higher percentage of forest cover in coastal Massachusetts than in inland region (green indicates forest remnants; white areas are open pastures or farms; gray area indicates no data).

Sample Collection and Genotyping

We collected leaf and bud tissue from 501 juvenile individuals from 28 forest stands with different histories throughout Massachusetts (Figure 1A). Our goal was to collect 30 individuals from each forest stand, leaving a 5-m gap between samples to reduce the sampling of clones. In forest stands where beech trees were few, we collected as many individuals as possible. We collected beech individuals in logged forests from 6 coastal or island sites (spanning a distance of 112 km) and 8 from western/central (inland) region (~108 km). We also sampled 6 and 8 recolonized forests in coastal and inland regions, respectively. In western and central Massachusetts, land-use history was determined by McDonald et al. (2008). For the coastal/island sites, we distinguished logged from recolonized forests based on a forest survey map, completed in 1830 near the peak of agriculture deforestation (Figure 1B) (Massachusetts Archives 1830, USCGS 1845-61).

Genomic DNA was extracted following the standard cetyltrimethyl ammonium bromide protocol with the addition of 1% polyvinylpyrrolidone and purification using phenol:chloroform:isoamyl alcohol (Doyle and Doyle 1987). Individuals were genotyped initially across 8 microsatellite loci. Seven of these markers (*FS1-03*, *FS1-11*, *FS1-15*, *FS3-04*, *FCM5*, *αCAT14*, and *mfc5*) were developed for other Fagaceae species (Tanaka et al. 1999; Marinoni et al. 2003; Pastorelli et al. 2003), whereas one marker, *P3566*, was derived from a subset of simple sequence repeat-containing expressed sequence tags (ESTs) identified in a 454-EST database for American beech (Staton et al. 2009). Amplifications were carried out in a 25-μL reaction containing 0.2 μM primer, 0.05 unit *Taq* polymerase (New England Biolabs), 400 μM dNTPs, 1.6 mM MgCl₂ and 1× polymerase chain reaction (PCR) buffer and 10–20 ng of template DNA. Cycling parameters include initial denaturation at 94 °C for 10 min, 35 cycles of 94 °C for 45 s, 53–57.6 °C for 45 s, 72 °C for 1 min, and final elongation of 72 °C for 10 min. PCR products were analyzed with an ABI Prism 3730 in the Genomics Core Facility at University of Notre Dame. Allele sizes were scored using GeneMapper or PeakScanner v.1 (Applied Biosystems). Ambiguous samples (40 out of 500 individuals) were amplified and genotyped again. We have deposited data used in all analyses with Dryad.

Testing Model Assumptions

Deviation from Hardy–Weinberg Equilibrium (HWE) and evidence of linkage disequilibrium (LD) were tested for all possible pairs of loci and populations using the HWE and LD probability test, respectively, in GENEPOP on the web (Raymond and Rousset 1995). Evidence of null alleles was checked using INEst (Chybicki and Burczyk 2009), which simultaneously estimates inbreeding coefficient and null alleles, using the Population Inbreeding Model (PIM) (jack-knife estimation). In further analysis, we excluded problematic loci with high frequency (>0.19) of null alleles from our data set. This value has been shown as a threshold over which significant underestimate of H_E due to null alleles can be found (Chapuis et al. 2008). Where possible, we employ

programs that are robust in presence of null alleles, particularly in estimation of F -statistics.

Genetic Diversity Analysis

We first estimated and compared genetic diversity in logged beech forests between coastal and inland region in order to determine baseline patterns of genetic variation before the historic forest clearance. Observed heterozygosity (H_O), allelic richness (A_R), and unbiased within-population gene diversity (H_S) (expected heterozygosity) were calculated per locus and across all loci and corrected for the lowest population size (i.e., 13) using FSTAT v.2.9.3.2 (Goudet 2001). We also computed the effective number of alleles (A_E) in SpaGeDi v1.4 (Hardy and Vekemans 2002), as well as private alleles (unique alleles, A_U) in Arlequin v3.5 (Excoffier et al. 2005). We compared the mean of each category for each of the genetic diversity parameter (i.e., parameters were estimated first for each population and then averaged for each category) using FSTAT v.2.9.3.2. Significance of comparison was assessed using the nonparametric tests with 10 000 permutations implemented in the program.

Next, we examined the genetic impacts of reforestation by comparing levels of genetic diversity between logged forests and recolonized forests: first, across our entire study site to examine overall genetic impacts of forest recolonization without regard to habitat differences (hypothesis 2), and second, between logged and recolonized forests within each coastal and inland group to account for the impact of habitat differences (hypothesis 3). All comparison tests were conducted between the group (population) mean of each category using FSTAT, with significance test evaluated with 10 000 permutations. We expect reduction of genetic diversity in recolonized forests compared with logged forests and that this pattern will be more pronounced in coastal region.

Population Genetic Structure and Differentiation

We first examined overall population structure and differentiation across our study region regardless of forest history. We computed global and pairwise F_{st} between populations using the program FreeNA (Chapuis and Estoup 2007) to examine the genetic differentiation across all populations. The Estimation of Null Allele method in FreeNA efficiently corrects for the positive bias on F_{st} estimation posed by the presence of null alleles. Overall population genetic structure was determined (regardless of region and forest history) without a prior assumption using the Bayesian clustering program BAPS version 5.3 (Corander et al. 2003). This allows us to assess the broadscale population structure across our entire study region. BAPS infer parameters of model consistent with multilocus genotypic data, treating them as variable and therefore explore a wide parameter space. It partitions data among groups based on similarities in population allele frequencies. We conducted BAPS analysis across all populations ($K = 28$) to determine overall structure using the clustering method, with 100 iterations. Migration and shared ancestry result in weaker structure among populations, whereas genetic drift and low gene flow result in stronger structure.

We further investigated the patterns of regional spatial genetic structure (SGS) of logged forests separately within inland and coastal regions to examine genetic structure prior to 19th-century forest clearance and therefore have a better context for assessing the impact of this forest clearance. We assessed significance of relationship between (regional) pairwise population F_{st} values and geographic distance (km) for each region separately using a Mantel test with 10 000 randomizations implemented in IBDWS (Jensen et al. 2005). This analysis also tests for isolation-by-distance (IBD) pattern among populations within each region, which can lead to spatial genetic structuring among populations as wind-pollinated species are capable of long-distance dispersal. Regional pairwise F_{st} and corresponding geographic distances were calculated in SpaGeDi v.1.4 (Hardy and Vekemans 2002). To examine whether degree of scatter increases with increasing geographic distance within each region, we calculated residuals from the standard linear regression of pairwise F_{st} values on corresponding geographical distances and then regressed this onto geographic distance and the significance was assessed, similar to de LaFontaine et al. (2013). We expect low degree of scatter if there is a significant relationship between genetic and geographic distance and hence a significant SGS pattern among populations as a consequence of IBD process. All statistical tests were carried out in R (R Development Core Team 2010). We also compared group mean F_{st} values of logged forests between inland versus coastal region using FSTAT to examine whether coastal populations show stronger genetic differentiation than inland populations (hypothesis 1).

Forest regeneration can alter the genetic structure and differentiation among the resulting recolonized populations. We first assessed whether recolonized populations show increased genetic divergence by comparing levels of genetic differentiation (F_{st}) between 1) logged versus recolonized forest and 2) logged versus recolonized forests within each inland and coastal groups. Comparison tests were conducted between group (population) mean of each category using FSTAT, with significance evaluated with 10 000 permutations. Next, we examined if recolonized forests maintained the patterns of SGS observed in logged forests within each region using similar SGS approach described above. A 2-way Anova test was conducted using R v.2.15.1 (R Development Core Team 2010) to examine more fully the interaction between fragmentation and environmental suitability for each measure of genetic diversity, using forest history (logged vs. secondary) and region (coastal vs. inland) as factors.

Historical Bottleneck and Inbreeding Levels

Founder events, such as colonization of new habitats by a small number of individuals, can lead to genetic bottlenecks. We used 3 tests to detect genetic bottlenecks. First, we calculated the M -ratio of each population using ARLEQUIN v.3.5. M -ratio is the mean ratio of actual number of alleles to range in all possible allele sizes. It detects populations that have undergone large reduction in effective population size at a critical value of 0.68 (Garza and Williamson 2001).

The other 2 tests detect more recent bottlenecks (Williamson-Natesan 2005; Girod et al. 2011). Allele frequency mode shift examines shifts in the typically skewed allele frequency distribution (i.e., “L” shape) spectrum of populations at equilibrium due to preferential loss of low-frequency alleles when population undergoes recent reduction in effective population size. Finally, we tested for heterozygote excess by comparing actual heterozygosity to that expected in a population under mutation-drift equilibrium. We examined both tests using a 1-tailed Wilcoxon signed-rank test in BOTTLENECK v.1.2.02 (Luikart and Cornuet 1998) under 2-phase mutation model of evolution (proportion of stepwise mutations = 0.70, $\sigma^2 = 30$), which fits microsatellite markers better.

We estimated inbreeding levels (F_{is}) using the program INEst (Chybicki and Burczyk 2009). We used the PIM, which calculates the population-wide average inbreeding coefficient, as our study is more concerned with overall inbreeding of a population than individual inbreeding coefficient.

Factors Explaining Genetic Variation

We investigated factors that could contribute to or affect our population genetic measures for each coastal and inland group, using methods similar to those used by Hoban et al. (2010). Sites with no historical data were excluded from the analysis. The primary variable of interest is land-use history. We determined the best predictors of the measures of genetic diversity (A_R , H_O , H_S , N_A , and E_A) using the following variables: 1) beech percentage at start of settlement as proxy for long-term historical beech abundance, 2) forest history (secondary or logged), and 3) mean size of forest fragments within 5 km of the site at the peak of agriculture deforestation. We estimated the relative abundance of beech in forests at the time of settlement based on settlement-era surveys. Mean fragment size was calculated as the average forest fragment size at the peak of agriculture deforestation (ca. 1830) surrounding each site. Multiple regression analysis was used for each of the 5 response variable (A_R , H_O , H_S , N_A , and E_A) using the 3 predictor variables. We performed model simplification using a backward removal of nonsignificant terms using R v.2.15.1 (R Development Core Team 2010). We performed these analyses separately for inland and coastal groups.

Results

Null Alleles and HWE

One locus, *FS1-03*, showed a frequency of null alleles >0.19 and was excluded from our data set. The presence of null alleles was also suspected in other loci with frequencies ranging from 0.03 (*ωCAT14*) to 0.10 (*FS1-15*), although they are randomly distributed across all populations. To determine if PCR and scoring errors or large allele dropout could have contributed to the observed heterozygote deficit, we resequenced a subset of individuals (35 individuals) that are homozygous at a particular locus (mostly *FS1-03* and *FS1-15*) across

populations 3 more times. With the exception of 5 samples where amplification of another allele at 1 locus was obtained, all individuals showed similar results.

The mean number of alleles per locus per population ranged from 4.28 to 13.143. The total number of alleles (N_A) across all 7 loci is 240 (mean N_A per locus = 34.2 and mean N_A per population = 8.57). We found minimal significant LD (11 out of 588 possible locus–population combination). Test of HWE per locus on each population revealed that 44 out of 196 combinations (22.4%) showed significant departures. As beech is prolific sprouter, we looked at individuals sharing the same genotypes and found only four with the same genotype.

Population Genetic Diversity and Differentiation in Logged Forests

Overall, logged beech populations exhibited moderate levels of genetic diversity across the landscape. The observed heterozygosity and gene diversity were 0.557 and 0.694, respectively, whereas the overall allelic richness (A_R) (averaged across all populations and loci) was 6.117 (Table 1). We observed a clear trend in genetic diversity levels across the landscape. Coastal populations showed significantly lower levels of allelic richness, within-population gene diversity, and effective number of alleles than inland populations, supporting our first hypothesis (Table 2 and Figure 2).

Genetic differentiation among logged coastal populations was not significantly different from that of inland populations although the level was almost twice as high in the coast (Table 2). We found no significant relationship between regional pairwise F_{st} and geographic distance in either inland ($r = -0.017$, $P = 0.466$) or coastal region ($r = -0.215$, $P = 0.627$). Furthermore, scatter plot between genetic and geographic distance indicates that the degree of scatter did not significantly increase with geographic distance in either region although it was higher in the coast than in the inland region (residual F_{st} variance coast = 0.001; inland = 0.0004). These patterns indicate no significant SGS within each region as a consequence of IBD.

Historical Genetic Bottlenecks and Inbreeding Levels

All populations have M -ratio values lower than the critical value 0.68, suggesting the presence of an older genetic bottleneck, likely associated with the postglacial migration of American beech (Table 1). None of the populations showed evidence of more recent genetic bottlenecks (based on mode shift in allele frequency spectrum), although 4 populations showed heterozygosity excess. Tests for inbreeding levels (positive F_{is}) indicate that 8 populations exhibit positive F_{is} values although only 2 populations (both recolonized forests from inland region) showed F_{is} levels significantly different from 0, suggesting inbreeding in these 2 populations (Table 1).

Population Genetic Diversity and Differentiation After European Settlement

Observed heterozygosity across our study area was significantly lower in all recolonized forests, supporting our second hypothesis (Table 3). Allelic richness, within-population gene diversity,

and effective number of alleles were lower, but not significantly so in recolonized forests. Among coastal forests, recolonized forests were not significantly less diverse than logged forests (Table 4). Within the inland group, only observed heterozygosity was significantly reduced in recolonized forests.

Genetic differentiation increased significantly among recolonized forests on the coast but not in inland forests (Table 4). Across all populations, the increase in differentiation was not significant (Table 3). BAPS clustering identified 9 beech population clusters across both logged and recolonized forests ($K = 9$), with 8 coastal clusters and 1 large cluster comprising all (logged and recolonized) inland populations along with one of the logged coastal populations (Figure 3). The overall population genetic differentiation among all beech populations was high, with mean pairwise $F_{st} = 0.110$ ($P < 0.05$). A weak but significant pattern of IBD (significant relationship between pairwise F_{st} and geographic distance) was observed across all recolonized populations ($r = 0.386$, $P < 0.05$) and among recolonized forests in the inland region ($r = 0.381$, $P = 0.04$) but not on the coast ($r = -0.1607$, $P = 0.60$). However, scatter plot indicates that the degree of scatter (residual F_{st} variance) was lower in recolonized inland compared with recolonized coastal populations (residual F_{st} variance coast = 0.007; inland = 0.0008), though it did not significantly increase with geographic distance in either region. The 2-way Anova test revealed no significant interaction between region and forest history, although both region and forest history have significant effects on all measures of genetic diversity.

Factors Explaining Genetic Variation

In the multiple regression analysis across all sites, none of the explanatory variables, land-use history, percent beech at time of settlement, or fragment size at the peak of agriculture, significantly explained A_R , H_s , H_o , or A_E (Table 5). However, variation in the number of alleles (N_A) was significantly explained by forest fragment size in coastal populations, whereas variation in observed heterozygosity (H_o) was significantly explained by land-use history in inland populations.

Discussion

The question of how much genetic variation is recovered when forests recolonize cleared agriculture lands is important to consider in the context of current and future land-use changes. We found that the extent of recovery of genetic variation in a reforestation landscape varies in populations at different habitats. In marginal coastal habitats, the process of recolonization of agriculture lands led to increased divergence among post-agriculture forests although no reduction in the overall genetic diversity. This was not the case in inland (core) region.

Prehistoric Context for the Impact of Deforestation and Recovery of Genetic Diversity

The most recent event shaping diversity in the forests of Massachusetts was the extensive agricultural deforestation following European settlement 400 years ago (Foster et al. 2002; Hall et al. 2004). However, these impacts play out against a

Table 1 Population information and genetic diversity estimated across all loci

Population ID	Forest history	Latitude	Longitude	Location	Number of samples	A _U	A _E	A _R	H _O	H _S	F _{IS}	M-ratio	Mode shift	Heterozygosity excess
1	Logged	42.519	-72.205	Inland	30	4	4.718	7.657	0.557	0.792	0	0.333	N	N
2	Logged	42.307	-72.577	Inland	27	2	3.724	7.336	0.538	0.735	0	0.493	N	N
3	Recolonized	42.494	-72.019	Inland	14	2	3.736	7.347	0.418	0.735	0.474*	0.429	N	N
4	Logged	42.539	-72.66	Inland	20	2	3.507	6.491	0.593	0.723	0.001	0.489	N	N
5	Recolonized	42.717	-72.274	Inland	13	4	4.187	6.369	0.516	0.766	0	0.365	N	N
6	Logged	42.224	-73.289	Inland	14	3	4.785	6.511	0.612	0.796	0	0.404	N	N
7	Recolonized	42.271	-72.93	Inland	20	2	4.909	6.637	0.393	0.811	0.450*	0.405	N	N
8	Recolonized	42.584	-72.681	Inland	14	0	3.871	6.079	0.48	0.747	0.070	0.463	N	N
9	Logged	42.595	-73.164	Inland	13	1	2.894	6.059	0.624	0.66	0	0.37	N	Y
10	Recolonized	42.39	-73.174	Inland	13	2	4.282	6.259	0.467	0.772	0	0.435	N	N
11	Recolonized	42.527	-72.938	Inland	27	2	3.303	6.384	0.586	0.706	0	0.451	N	N
12	Recolonized	42.226	-72.75	Inland	14	3	3.772	6.85	0.582	0.746	0	0.32	N	N
13	Logged	42.17	-72.989	Inland	20	2	3.863	7.677	0.6	0.747	0.065	0.432	N	N
14	Logged	42.496	-73.188	Inland	14	4	3.332	6.789	0.561	0.705	0	0.36	N	Y
15	Logged	42.673	-73.153	Inland	14	2	4.546	6.661	0.582	0.79	0.175	0.287	N	Y
16	Recolonized	42.704	-72.188	Inland	14	0	3.966	6.748	0.473	0.759	0.022	0.375	N	N
17	Logged	42.071	-70.2	Coastal	24	1	2.784	5.613	0.649	0.641	0	0.305	N	N
18	Recolonized	41.963	-70.017	Coastal	14	1	3.131	5.014	0.714	0.679	0	0.303	N	N
19	Logged	41.556	-71.26	Coastal	24	2	2.47	5.539	0.649	0.594	0	0.336	N	N
20	Logged	41.429	-70.697	Coastal	14	1	2.988	4.457	0.548	0.574	0	0.238	N	N
21	Recolonized	41.434	-70.694	Coastal	14	0	2.377	3.892	0.405	0.502	0.010	0.348	N	N
22	Recolonized	41.469	-70.768	Coastal	15	2	2.98	5.86	0.606	0.667	0	0.342	N	N
23	Logged	41.497	-70.726	Coastal	16	0	2.681	6.341	0.58	0.629	0	0.287	N	Y
24	Logged	41.534	-70.659	Coastal	23	0	3.462	6.441	0.609	0.713	0	0.545	N	N
25	Recolonized	41.57	-70.61	Coastal	16	1	2.708	5.623	0.545	0.634	0	0.382	N	N
26	Recolonized	41.766	-70.028	Coastal	20	0	2.015	3.292	0.629	0.5	0.145	0.35	N	N
27	Recolonized	41.491	-70.733	Coastal	20	1	2.554	5.112	0.479	0.612	0	0.341	N	N
28	Logged	41.668	-70.478	Coastal	20	4	3.303	6.244	0.607	0.7	0	0.456	N	N
	Mean				18	2	3.459	6.117	0.557	0.694				

template formed by the earlier, postglacial history of species (Petit et al. 1997). Phylogeographic studies of American beech have shown that Massachusetts was colonized by a single beech chloroplast lineage from southern populations ~10 000 years ago (McLachlan et al. 2005; Morris et al. 2010). Across our entire study site, all logged beech populations exhibited historical genetic bottlenecks, consistent with founder events typical of postglacial colonization. The overall moderate levels of nuclear genetic diversity throughout the region mirror the low beech chloroplast DNA diversity

Table 2 Population genetic structure and genetic diversity of logged forests compared between inland and coastal regions

Genetic parameters	Inland	Coastal	P value
Allelic richness, A_R	6.898	5.772	0.030*
Observed heterozygosity, H_O	0.579	0.604	0.560
Within-population gene diversity, H_S	0.766	0.646	0.006*
Effective number of alleles, E_A	7.670	5.637	0.013*
Private alleles, A_U	2.625	1.333	0.053
Genetic differentiation, F_{st}	0.039	0.072	0.550

Values are the group mean for each category.

*Significant values.

found in Massachusetts from these earlier range-wide studies (McLachlan et al. 2005; Morris et al. 2010).

Subsequent influence of the environment on beech populations after their colonization and establishment led to a marked difference in levels and patterns of genetic variation between the inland (core) and peripheral coastal populations. The reduced genetic diversity and strong genetic differentiation we found among logged coastal populations in contrast to inland populations suggest that coastal populations experienced genetic isolation after their postglacial arrival, due to marginality of the coastal environment for beech. Environmental marginality influences the degree of spatial isolation in suitable patches of habitats, leading to genetic isolation of populations although gene flow might still occur among neighboring populations under such geographically restricted dispersal (Eckert et al. 2008). As a consequence, strong SGS pattern can be generated due to IBD process (although other factors might cause SGS as well) (Hutchison and Templeton 1999; de LaFontaine et al. 2013). In our study, the lack of SGS and the individual clusters identified in BAPS analysis among coastal populations indicate that IBD was not a predominant force in shaping genetic structure among populations. The historic isolation, combined with small population size, may have caused these coastal beech

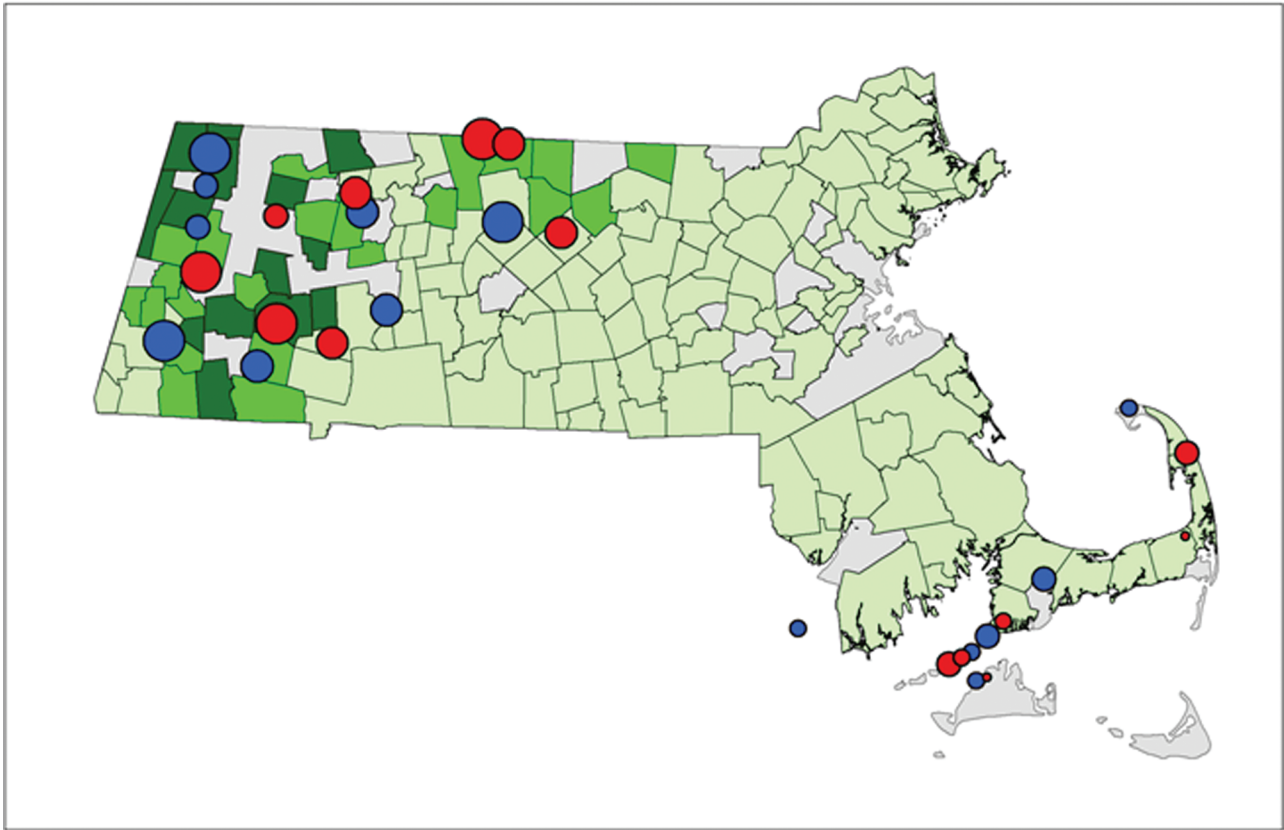


Figure 2. Inland populations, where beech was more abundant in settlement-era forests (same as Figure 1), show higher within-population gene diversity, H_S , than coastal populations (smaller circles). Red circles are recolonized forests, whereas blue circles are logged forests. The size of circle is proportional to H_S values.

Table 3 Population genetic estimates compared between logged and recolonized forests

Genetic parameters	Land use		P value
	Logged forests	Recolonized forests	
Allelic richness, A_R	6.415	5.819	0.130
Observed heterozygosity, H_O	0.590	0.520	0.010*
Within-population gene diversity, H_S	0.712	0.674	0.250
Effective number of alleles, E_A	6.799	5.977	0.241
Private alleles, A_U	2.071	1.357	0.149
Genetic differentiation, F_{st}	0.080	0.139	0.119

Values reported are the group mean for each category.

*Significant values.

Table 4 Population genetic structure and genetic diversity average across all loci compared between logged and recolonized forests in coastal and inland groups

Genetic parameters	Logged forests	Recolonized forests	P value
Coastal			
Allelic richness, A_R	5.772	4.799	0.100
Within-population gene diversity, H_S	0.646	0.594	0.310
Observed heterozygosity, H_O	0.604	0.553	0.280
Effective number of alleles, E_A	5.637	4.235	0.077
Private alleles, A_U	1.333	0.833	0.483
Genetic differentiation, F_{st}	0.072	0.249	0.001*
Inland			
Allelic richness, A_R	6.898	6.584	0.550
Within-population gene diversity, H_S	0.766	0.736	0.500
Observed heterozygosity, H_O	0.579	0.494	0.030*
Effective number of alleles, E_A	7.670	7.284	0.564
Private alleles, A_U	2.625	1.75	0.159
Genetic differentiation, F_{st}	0.039	0.054	0.780

Values reported are the group mean for each category.

*Significant values.

populations to diverge independently to the extent that IBD is no longer taking place and to lose genetic variation over time via genetic drift. Our results also suggest that the isolation of beech populations in presettlement forests at the coast is the predominant force in shaping their genetic structure, overcoming their effective gene dispersal through seeds and pollen.

Genetic Impact of Deforestation

Logging can potentially have negative genetic impacts on tree populations as a consequence of removing reproductive individuals in a population (e.g., [Baucom et al. 2005](#)). We did not find evidence of recent genetic bottlenecks in any of the remnant beech forests, which were all likely repeatedly logged. The brief bout of extensive regional deforestation in eastern North America (a one-generation disturbance for beech) apparently did not disrupt the breeding system of American beech to the extent observed in European beech over a longer period of forest fragmentation ([Jump and Penuelas 2006](#)). We note that beech is a prolific sprouter, so it is likely that many original trees survived repeated logging episodes in logged populations. We also note though that

this disturbance might be too recent (relative to life span of beech) for genetic bottlenecks to be detected.

Recovery of Genetic Variation Across a Gradient of Environmental Suitability

Recolonized forests, by contrast, must originate from these logged (remnant) populations, and postagricultural reforestation in our recolonized populations appears to have had an impact on the recovery of genetic variation. Across our entire study site, we found an overall decrease in heterozygosity but no increase in genetic differentiation in recolonized forests compared with logged forests. In marginal coastal habitat, the more pronounced genetic structure in logged coastal populations became further exacerbated as genetic differentiation increased in these newly recolonized forests ([Table 4](#)). Moreover, similar to logged coastal populations, we observed strong population structure in recolonized forests based on BAPS analysis, but this was not correlated with distance (no significant SGS). The lack of SGS in recolonized populations might be due to recent establishment of these populations, thus no sufficient time for dispersal-drift equilibrium to occur ([Hutchison and Templeton 1999](#)). These results indicate that

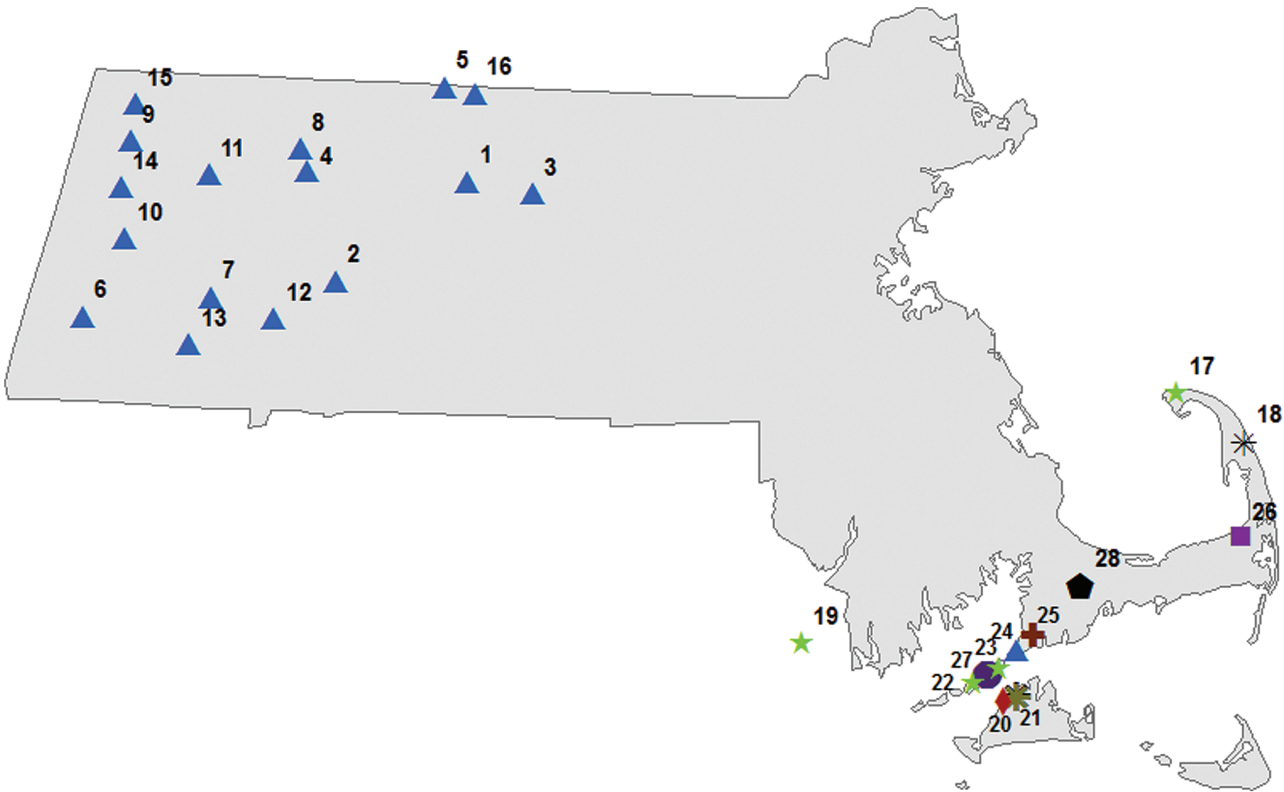


Figure 3. Genetic clustering of beech populations across Massachusetts based on BAPS. All inland populations are grouped together in a single, possibly panmictic, population (triangle). On the other hand, coastal populations show strong genetic differentiation.

Table 5 Factors explaining genetic variation in coastal and inland groups: probabilities for each predictor variables based on multiple linear regression tests

Response variable	Forest type		Percent beech		Mean 1830 forest fragment size	
	Coastal	Inland	Coastal	Inland	Coastal	Inland
Allele number, N_A	ns	ns	ns	ns	0.02*	ns
Effective number of alleles, A_E	ns	ns	ns	ns	ns	ns
Allelic richness, A_R	ns	ns	ns	ns	ns	ns
Observed heterozygosity, H_O	ns	0.003*	ns	ns	ns	ns
Gene diversity, H_S	ns	ns	ns	ns	ns	ns

ns, not significant. *Significant values.

the processes shaping genetic structure in logged forests at marginal habitat are accentuated during postagricultural recolonization. Genetic differentiation may increase or decrease in recolonized populations depending on both the number of founding individuals and subsequent migrants (Wade and McCauley 1988). Our results suggest that only relatively few founding individuals of local origin recolonized agriculture lands, possibly due to the marginality of coastal environment that limits seed dispersal and seedling establishment. Moreover, these founders apparently do not represent the breadth of genetic variation found in logged forests in the coast. Recolonized forests in the coast received only a subset of local genetic sources at the time of colonization instead of

receiving mixed gene flow from differentiated logged populations across the landscape, thus becoming more differentiated. This is similar to other reported findings (Vellend 2004; Baucom et al. 2005; Davies et al. 2010). Our results suggest that this strong genetic differentiation may persist into later generations given the limited gene flow (seed- and pollen-mediated gene flow) among beech populations on the coast.

The recovery of genetic diversity in recolonized populations in inland region, on the other hand, appears more complete than on the coast. Overall genetic diversity in inland recolonized forests was not significantly lower than that in logged forests. Moreover, the original genetic structure of the logged forests was maintained as all inland populations

regardless of forest history belong to a single panmictic population, with no significant increase in genetic differentiation in recolonized forests (Table 4). This overall low impact of deforestation in inland populations is presumably due to the presence of genetically diverse and relatively large source populations. Such immediate recovery of genetic diversity in recolonized populations can occur if the founding individuals come from genetically diverse sources and high gene flow from these sources was subsequently maintained. High levels of gene flow among populations have been shown to balance the effect of fragmentation or delay the loss of genetic diversity (Fore et al. 1992; Craft and Ashley 2010; Leonardi et al. 2012). Maintenance of genetic diversity in recolonized forests had also been reported in another study of American beech in a different part of its range (Kitamura et al. 2008).

Conservation Implications

The potential of human land-use changes to significantly impact tree population genetics is still a subject of continuing research (Hamrick 2004; Kramer et al. 2008). Although temperate forests are generally considered resistant to the negative effects of fragmentation (Hamrick 2004), our findings illustrate the importance of taking into account differences in habitat as they can influence the extent of genetic recovery in recolonized forests. In marginal coastal habitat, the factors that alter the level of genetic differentiation and genetic variation in natural forests are amplified by even a short bout of regional disturbance. This finding is relevant to conservation genetic planning in the context of shifting trends in land use and also in the context of population shifts under climate change. For instance, available models suggest that New England's climate will become less favorable for American beech this century (Prasad et al. 2007) and this might be true for other species as well. If anthropogenic pressures continue to increase amidst changing climate that results in less favorable environment, the genetic consequences of fragmentation typical of the current forests would then become more severe in the future. However, our results also indicate that these negative impacts can be ameliorated given the extensive gene flow among populations in favorable habitats. More generally, our findings are also relevant in informing conservation efforts such as assisted migration and reforestation whose aims are to maximize genetic diversity within and among populations. Understanding the process of recovery and development of genetic diversity in newly recolonized forests, for instance, can give insights into how a given level of genetic diversity within populations can be maintained (Frankham 2010).

Funding

Notre Dame Center for Aquatic Conservation (CAC).

Acknowledgments

We thank the following people for their contributions: P. Busby, J. Lozano, G. Motzkin, J. Mikels-Carrasco, and J. Deines. We are also grateful to

C. Pachico, K. Binder, and C. Rodstrom (Seven Gates Farm) for permission to collect beech samples and Harvard Forest for the forest data. This article greatly benefited from comments of S. Hoban, L. Anderson-Carpenter, and the McLachlan Lab Group.

References

- Baucom RS, Estill JC, Cruzan MC. 2005. The effect of deforestation on the genetic diversity and structure in *Acer saccharum* (Marsh): evidence for the loss and restructuring of genetic variation in a natural system. *Conserv Genet.* 6:39–50.
- Busby PE. 2006. American beech in coastal New England: forest history and dynamics [MSc thesis]. [Cambridge (MA)]: Harvard University. 123 p.
- Busby PE, Motzkin G, Hall B. 2009. Distribution and dynamics of American beech in southern New England. *Northeast Nat.* 16:159–176.
- Canham CD. 1990. Suppression and release during canopy recruitment in *Fagus grandifolia*. *Bull Torrey Bot Club.* 117:1–7.
- Céspedes M, Gutierrez MV, Holbrook NM, Rocha OJ. 2003. Restoration of genetic diversity in the dry forest tree *Swietenia macrophylla* (Meliaceae) after pasture abandonment in Costa Rica. *Mol Ecol.* 12:3201–3212.
- Chapuis MP, Estoup A. 2007. Microsatellite null alleles and estimation of population differentiation. *Mol Biol Evol.* 24:621–631.
- Chapuis MP, Lecoq M, Michalakakis Y, Loiseau A, Sword GA, Piry S, Estoup A. 2008. Do outbreaks affect genetic population structure? A worldwide survey in *Locusta migratoria*, a pest plagued by microsatellite null alleles. *Mol Ecol.* 17:3640–3653.
- Chauchard S, Carcaillat C, Guibal F. 2007. Patterns of land-use abandonment control tree-recruitment and forest dynamics in Mediterranean mountains. *Ecosystems.* 10:936–948.
- Chybicki IJ, Burczyk J. 2009. Simultaneous estimation of null alleles and inbreeding coefficients. *J Hered.* 100:106–113.
- Cogbill C, Burke J, Motzkin G. 2002. The forest of presettlement New England USA: spatial and compositional patterns based on proprietor surveys. *J Biogeogr.* 29:1279–1304.
- Corander J, Waldmann P, Sillanpää MJ. 2003. Bayesian analysis of genetic differentiation between populations. *Genetics.* 163:367–374.
- Craft KJ, Ashley MV. 2010. Pollen-mediated gene flow in isolated and continuous stands of bur oak, *Quercus macrocarpa* (Fagaceae). *Am J Bot.* 97:1999–2006.
- Davies S, Cavers S, Finnigan B, Navarro C, Lowe A. 2010. Genetic consequences of multigenerational and landscape colonisation bottlenecks for a neotropical forest pioneer tree, *Vochysia ferruginea*. *Trop Plant Biol.* 3:14–27.
- Davis MB. 1981. Quaternary history and the stability of forest communities. In: West DG, Shugart HH, Botkin DB, editors. *Forest succession: concepts and application*. New York: Springer-Verlag. p. 132–153.
- de Lafontaine G, Ducouso A, Lefèvre S, Magnanou E, Petit RJ. 2013. Stronger spatial genetic structure in recolonized areas than in refugia in the European beech. *Mol Ecol.* 22:4397–4412.
- Doyle JJ, Doyle JL. 1987. A rapid DNA isolation procedure from small quantities of fresh leaf tissues. *Phytochem Bull.* 19:11–15.
- Dubreuil M, Riba M, González-Martínez SC, Vendramin GG, Sebastiani F, Mayol M. 2010. Genetic effects of chronic habitat fragmentation revisited: strong genetic structure in a temperate tree, *Taxus baccata* (Taxaceae), with great dispersal capability. *Am J Bot.* 97:303–310.
- Eberhardt RW, Foster DR, Motzkin G, Hall B. 2003. Conservation of changing landscapes: vegetation and land-use history of Cape Cod National Seashore. *Ecol Appl.* 13:68–84.
- Eckert CG, Samis KE, Lougheed SC. 2008. Genetic variation across species' geographical ranges: the central-marginal hypothesis and beyond. *Mol Ecol.* 17:1170–1188.

- Excoffier L, Laval G, Schneider S. 2005. Arlequin (version 3.0): an integrated software package for population genetics data analysis. *Evol Bioinform Online*. 1:47–50.
- Fore SA, Hickley RJ, Vankat JL, Guttman SI, Schaefer RL. 1992. Genetic structure after forest fragmentation: a landscape ecology perspective on *Acer saccharum*. *Can J Bot*. 70:1659–1668.
- Foster DR. 1992. Land-use history (1730–1990) and vegetation dynamics in central New England, USA. *J Ecol*. 80:753–772.
- Foster DR. 2002. The physical and biological setting for ecological studies. In: Foster DR, Aber JD, editors. *Forests in time: the environmental consequences of 1000 years of change*. New Haven (CT): Yale University Press. p. 19–31.
- Foster DR, Motzkin G, Okeefe J, Boose E, Orwig D, Fuller J, Hall B. 2002. The environmental and human history of New England. In: Foster DR, Aber JD, editors. *Forests in time: the environmental consequences of 1,000 years of change in New England*. New Haven (CT): Yale University Press. p. 43–142.
- Foster DR, Motzkin G, Slater B. 1998. Land-use history as long-term broad-scale disturbance: regional forest dynamics in central New England. *Ecosystems*. 1:96–111.
- Frankham R. 2010. Challenges and opportunities of genetic approaches to biological conservation. *Biol Conserv*. 143:1919–1927.
- Fuller JL, Foster DR, McLachlan JS, Drake N. 1998. Impact of human activity on regional forest composition and dynamics in central New England. *Ecosystems*. 1:76–95.
- Fuller J, Foster DR, Motzkin G, McLachlan J, Barry S. 2002. Broad-scale forest response to land use and climate change. In: Foster DR, Aber JD, editors. *Forests in time: the environmental consequences of 1,000 years of change in New England*. New Haven (CT): Yale University Press. p. 101–124.
- Garza JC, Williamson EG. 2001. Detection of reduction in population size using data from microsatellite loci. *Mol Ecol*. 10:305–318.
- Gillman M. 1997. Plant population ecology. In: Maxted N, Ford-Lloyd BV, Hawkes JG, editors. *Plant genetic conservation: the in situ approach*. London: Chapman & Hall. p. 114–131.
- Girod C, Vitalis R, Leblois R, Fréville H. 2011. Inferring population decline and expansion from microsatellite data: a simulation-based evaluation of the Msvr method. *Genetics*. 188:165–179.
- Goudet J. 2001. FSTAT, a program to estimate and test gene diversities and fixation indices (version 2.9.3). [Cited 2012 May 12]. Available from: www.unil.ch/izea/software/fstat.html.
- Hall B, Motzkin G, Foster DR, Syfert M, Burk J. 2004. Three hundred years of forest and land-use change in Massachusetts, USA. *J Biogeogr*. 29:1319–1335.
- Hamrick JL. 2004. Response of forest trees to global environmental changes. *For Ecol Manage*. 197:323–335.
- Hardy OJ, Vekemans X. 2002. SPAGeDi: a versatile computer program to analyse spatial genetic structure at the individual or population levels. *Mol Ecol Notes*. 2:618–620.
- Hewitt GM. 2004. The structure of biodiversity—insights from molecular phylogeography. *Front Zool*. 1:4.
- Hoban SM, Borkowski DS, Brosi SL, McCleary TS, Thompson LM, McLachlan JS, Pereira MA, Schlarbaum SE, Romero-Severson J. 2010. Range-wide distribution of genetic diversity in the North American tree *Juglans cinerea*: a product of range shifts, not ecological marginality or recent population decline. *Mol Ecol*. 19:4876–4891.
- Houston DR, Parker EJ, Lonsdale D. 1979. Beech bark disease: patterns of spread and development of the initiating agent *Cryptococcus fagisuga*. *Can J For Res*. 9:336–344.
- Hutchison DW, Templeton AR. 1999. Correlation of pairwise genetic and geographic distance measures: inferring the relative influences of gene flow and drift on the distribution of genetic variability. *Evolution*. 53:1898–1914.
- Jensen JL, Bohonak AJ, Kelley ST. 2005. Isolation by distance, web service. *BMC Genet*. 6:13.
- Johnson WC, Adkisson CS. 1985. Dispersal of beechnuts by bluejays in fragmented landscape. *Am Midl Nat*. 113:319–324.
- Jump AS, Penuelas J. 2006. Genetic effects of chronic habitat fragmentation in a wind-pollinated tree. *Proc Natl Acad Sci USA*. 103:8096–8100.
- Kitamura K, Kawano S. 2001. Regional differentiation in genetic components for the American Beech, *Fagus grandifolia* Ehrh., in relation to geological history and mode of reproduction. *J Plant Res*. 114:353–368.
- Kitamura K, Takasu H, Hagiwara S, Homma K, O'Neill J, Whigham DF, Kawano S. 2008. Demographic genetics of American beech (*Fagus grandifolia* Ehrh.) IV. Development of genetic variability and gene flow during succession in a coastal plain forest in Maryland. *Plant Spec Biol*. 23:159–173.
- Kramer AT, Ison JL, Ashley MV, Howe HF. 2008. The paradox of forest fragmentation genetics. *Conserv Biol*. 22:878–885.
- Lammi A, Siikamäki P, Mustajärvi K. 1999. Genetic diversity, population size, and fitness in central and peripheral populations of a rare plant *Lychnis viscaria*. *Conserv Biol*. 13:1069–1078.
- Leahy M, Pregitzer K. 2003. A comparison of presettlement and present-day forests in Northeastern Lower Michigan. *Am Midl Nat*. 149:71–89.
- Leimu R, Mutikainen P, Koricheva J, Fischer M. 2006. How general are positive relationships between plant population size, fitness and genetic variation? *J Ecol*. 94: 942–952.
- Leonardi S, Piovani P, Scalfi M, Piotti A, Giannini R, Menozzi P. 2012. Effect of habitat fragmentation on the genetic diversity and structure of peripheral populations of beech in Central Italy. *J Hered*. 103:408–417.
- Lesica P, Allendorf FW. 1995. When are peripheral populations valuable for conservation? *Conserv Biol*. 9:753–760.
- Lesser MR, Parchman TL, Jackson ST. 2013. Development of genetic diversity, differentiation and structure over 500 years in four ponderosa pine populations. *Mol Ecol*. 22:2640–2652.
- Luikart G, Cornuet J-M. 1998. Empirical evaluation of a test for identifying recently bottlenecked populations from allele frequency data. *Conserv Biol*. 12:228–237.
- Marinoni D, Akkac A, Bounous G, Edwards KJ, Botta R. 2003. Development and characterization of microsatellite markers in *Castanea sativa* (Mill.). *Mol Breed*. 11:127–136.
- Massachusetts Archives. 1830. 1830 Map series. Boston (MA): Massachusetts Archives.
- McDonald R, Motzkin G, Foster DR. 2008. The effect of logging on vegetation composition of Western Massachusetts. *For Ecol Manage*. 255:4021–4031.
- McLachlan JS, Clark J, Manos P. 2005. Molecular indicators of tree migration capacity under rapid climate change. *Ecology*. 86:2088–2098.
- Morris A, Graham C, Soltis D, Soltis P. 2010. Reassessment of phylogeographical structure in an eastern North American tree using Monmonier's algorithm and ecological niche modeling. *J Biogeogr*. 37:1657–1667.
- Ouborg NJ, Piquot J, van Groenendael JM, van Groenendael JM. 1999. Population genetics, molecular markers and the study of dispersal in plants. *J Ecol*. 87:551–568.
- Pastorelli R, Smulders MJ, van't Westende WPC, Vosman B, Giannini R, Vetorri C, Vendramin GG. 2003. Characterization of microsatellite markers in *Fagus sylvatica* L. and *Fagus orientalis* Lipsky. *Mol Ecol Notes*. 3:76–78.
- Petit RJ, Pineau E, Demesure B, Bacilieri R, Ducousso A, Kremer A. 1997. Chloroplast DNA footprints of postglacial recolonization by oaks. *Proc Natl Acad Sci USA*. 94:9996–10001.
- Poulson TL, Platt WJ. 1989. Gap light regimes influence canopy tree diversity. *Ecology*. 70:553–555.
- Prasad AM, Iverson LR, Matthews S, Peters M. 2007. A climate change Atlas for 134 forest tree species of the Eastern United States [cited 2012 Jan 27].

- Delaware (OH): Northern Research Station, USDA Forest Service. Available from: <http://www.nrs.fs.fed.us/atlas/tree>.
- Prishchepov AV, Muller D, Dubinin M, Baumann M, Radeloff CV. 2013. Determinants of agricultural land abandonment in post-Soviet European Russia. *Land Use Policy*. 30:873–884.
- Raymond M, Rousset F. 1995. GENEPOP (version 1.2): population genetics software for exact tests. *J Hered*. 86:248–249.
- R Development Core Team. 2010. R: a language and environment for statistical computing. Vienna (Austria): R Foundation for Statistical Computing. [Cited 2012 March 3]. Available from: <http://www.R-project.org>.
- Sexton JP, McIntyre PJ, Angert AL, Rice KJ. 2009. Evolution and ecology of species range limits. *Annu Rev Ecol Evol Syst*. 40:415–436.
- Sezen UU, Chazdon RL, Holsinger KE. 2007. Multigenerational genetic analysis of tropical secondary regeneration in a canopy palm. *Ecology*. 88:3065–3075.
- Sork VL, Nason J, Campbell DR, Fernandez JF. 1999. Landscape approaches to historical and contemporary gene flow in plants. *Trends Ecol Evol*. 14:219–224.
- Staton M, Barakat A, Ficklin S, Tompkins J, DiLoreto S, Carlson J. 2009. Genomic tool development for the Fagaceae. *Plant and Animal Genomes XVII Conference*; 2009 January 10–14; San Diego, CA. p. W190.
- Tanaka K, Tsumura Y, Nakamura T. 1999. Development and polymorphism of microsatellite markers for *Fagus crenata* and the closely related species, *F. japonica*. *Theor Appl Genet*. 99:11–15.
- Vellend M. 2004. Parallel effects of land-use history on species diversity and genetic diversity of forest herbs. *Ecology*. 85:3043–3055.
- Wade MJ, McCauley DE. 1988. Extinction and re-colonization: their effects on the genetic differentiation of local populations. *Evolution*. 42:995–1005.
- Ward RT. 1961. Some aspect of regeneration in American beech. *Ecology*. 42:828–832.
- Webb S. 1987. Beech range extension and vegetation history: pollen stratigraphy of two Wisconsin Lakes. *Ecology*. 68:1993–2005.
- Williamson-Natesan E. 2005. Comparison of methods for detecting bottlenecks from microsatellite loci. *Conserv Genet*. 6:551–562.

Received October 22, 2013; First decision January 16, 2014;
Accepted June 18, 2014

Corresponding Editor: David B. Wagner